
IK 510: Cognitive Neuroscience

Course Code : IK 510

Course Name : Cognitive Neuroscience

L-T-P-C : 3-1-0-4

Intended for : 3rd & 4th Year B. Tech, Masters and PhD

Prerequisite :

Mutual Exclusion:

Approval: 54th BoA

Course Contents

- **Structure and function of nervous system (5 hours)**
 - Overview of nervous system
 - Brain & spinal cord anatomy
 - Cells of nervous system
 - Action potential
 - Neurotransmitters
 - Synapse
 - Networks
- **Sensation & Perception (6 hours)**
 - Olfaction
 - Gustation
 - Somatosensation
 - Audition
 - Vision
 - Sensation to perception
- **Attention (4 hours)**
 - Models of attention
 - Neural mechanism of attention
- **Attention control networks Action (4 hours)**
 - Anatomy & control of motor structures
 - Motor pathways & physiology
 - Motor planning, initiation and control
 - Motor learning and memory
- **Memory (4 hours)**
 - Learning and memory-anatomical & cellular basis

- Mechanisms of memory
- Memory consolidation
- Memory deficits and amnesia
- **Emotions** (4 hours)
 - Defining & categorizing emotions
 - Theories of emotion generation
 - Neural systems of emotion processing
 - Emotion & cognition (learning & memory)
- **Language** (4 hours)
 - Evolution & fundamentals of language
 - Sensory, motor and cognitive aspects of language
 - Theories of language comprehension & speech production
 - Language deficits
- **Cognitive control** (4 hours)
 - Cognitive control anatomy, cross talks (sensorimotor) & deficits
 - Planning, Decision making & execution mechanisms
 - Cognitive control & neuropsychiatry
- **Social Cognition** (4 hours)
 - Theories of self
 - Understanding mental states of others (Theory of Mind-ToM)
 - Experience sharing & simulation theory
 - Mental state attribution
 - Neurobiology of social cognition
 - Social cognition deficits
- **Consciousness** (3 hours)
 - Mind-Brain problem
 - Consciousness theories- neuroscience perspective
 - Consciousness theories & practice-Yogic perspective
 - Miscellaneous-Plant & animal consciousness and others
 - Disorders of consciousness

Tutorials:

- Methods in cognitive neuroscience
 - Behavioral experiments-tools & techniques
 - Introduction to neurophenomenology
 - Basics of bio signals
 - Electroencephalograph (EEG) & Heart rate variability (HRV)
 - Transcranial electrical current stimulation(tECS)
 - Transcranial magnetic stimulation (TMS)
 - Functional neuroimaging techniques (fNIRS & fMRI)
 - Introduction to computational neuroscience (ML & AI)

Textbooks:

1. Gazzaniga, Michael S., et al., **Cognitive Neuroscience: The Biology of the Mind**, W.W. Norton, 2018.

References:

1. Kandel ER, Koester JD, Mack SH, Siegelbaum SA, **Principles of Neural Science**, 6th Edition, McGraw Hill, 2021